

10.3 Other Clustering Algorithms

When k-means is the wrong shape.

These notes walk through the lecture slides. The original deck is unchanged; use **(slides)** on the course hub if you want that PDF.

Figures and notes follow Géron, *Hands-On Machine Learning*; Deisenroth, Faisal, and Ong, *Mathematics for Machine Learning*; and Theodoridis.

1. Spectral clustering

Cluster by **similarity** of feature representations, not by Euclidean balls in the original space.

1. **Affinity matrix.** Pairwise similarities or distances. Points are vertices; edge weights are similarities.
2. **Graph Laplacian.** Turn the affinity matrix into a Laplacian. That matrix encodes connectivity.
3. **Eigendecomposition.** Eigenvectors of the Laplacian embed the points in a lower-dimensional space.
4. **Cluster the embedding.** Take the eigenvectors for the **smallest** eigenvalues. Run clustering (usually *k*-means) there.

Benefits: irregular shapes, non-convex clusters, nonlinear geometry. A flexible route on high-dimensional data.

Applications: image segmentation, social networks, gene expression, dimensionality reduction — anywhere the clumps are not round in the raw features.

2. The linear algebra

Pairwise similarity or distance. Euclidean distance of x_i and x_j :

$$d(x_i, x_j) = \sqrt{\sum_{k=1}^n (x_{ik} - x_{jk})^2}.$$

Cosine similarity of u and v :

$$\frac{u \cdot v}{\|u\| \|v\|}.$$

Affinity matrix A : entry a_{ij} is the similarity (or distance) of x_i and x_j .

Graph Laplacian $L = D - A$, where D is diagonal with the **degree** of each vertex.

Then eigendecompose L . Those eigenvectors are the embedding.

3. Computing eigenpairs

Family	Idea
Direct	QR, Jacobi, bisection. Straightforward; expensive on large matrices.
Iterative	Power iteration, Lanczos. Better for large sparse L .
Matrix decompositions	SVD or Cholesky, used indirectly.
Graph partitioning	Spectral partitioning or bisection: split the graph using eigenvectors, then work on the pieces.
Sparse libraries	ARPACK and friends: a few eigenpairs of a large sparse Laplacian.

Practice

1. Spectral clustering still runs k -means. In what space?
2. Write the graph Laplacian in terms of the affinity matrix A .
3. Why might you build A from cosine similarity instead of Euclidean distance?